

Section 1.1: Deductive VS Inductive Reasoning

Finite Mathematics MTH 107 Fall 2014

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Mars Hill University

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Deductive Reasoning

Reasoning from statements/assumptions to reach a logically *certain* conclusion.

- Applying a general rule to a specific instance.

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Example Argument

1. All men are mortal.
2. Socrates is a man.

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1. All men are mortal.
2. Socrates is a man.

Therefore, Socrates is mortal.

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Reasoning from statements/assumptions to reach a logically *certain* conclusion.

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- *Correct* premises give correct conclusions.

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1. The area of a triangle is $\frac{1}{2}b \cdot h$.

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1. The area of a triangle is $\frac{1}{2}b \cdot h$.
2. Triangle T has a base of 3 and height of 4.

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Example Argument

1. The area of a triangle is $\frac{1}{2}b \cdot h$.
 2. Triangle T has a base of 3 and height of 4.
- ⇒ Therefore, the area of T is 6.

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Inductive Reasoning

Reasoning in which the premises supply *strong evidence* for a conclusion.

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Example Argument

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Example Argument

- ① Socrates drank milk and got sick.
- ② Socrates ate ice cream and got sick.

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Reasoning in which the premises supply *strong evidence* for a conclusion.

- Using specific cases to make a general statement.
- Conclusion is *not* necessarily correct.

Example Argument

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 2. Triangle T has a base of 3 and height of 4.
- ⇒ Therefore, the area of T is 6.

Example Argument

- ① Socrates drank milk and got sick.
 - ② Socrates ate ice cream and got sick.
- ⇒ Therefore, Socrates is lactose intolerant.

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Example

1. All known life requires water.

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- ⇒ Therefore, the area of T is 6.

Example

1. All known life requires water.
Therefore, life will only be found on planets with water.

Valid/Invalid Arguments

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Invalid Argument

A deductive argument is *invalid* if the premises are met, but the conclusion does *not necessarily* follow.

- Use a Venn diagram.

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Example

1. All presidents are men.

Valid/Invalid Arguments

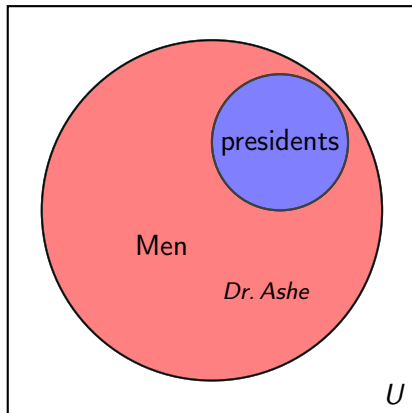
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A deductive argument is *invalid* if the premises are met, but the conclusion does *not necessarily* follow.

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Example

1. All presidents are men.
2. Dr. Ashe is a man.



Valid/Invalid Arguments

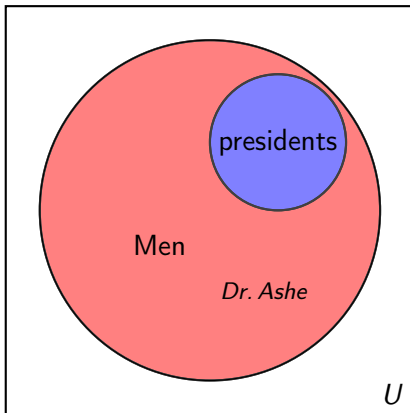
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1. All presidents are men.
 2. Dr. Ashe is a man.
- ⇒ Therefore, Dr. Ashe is a US president.



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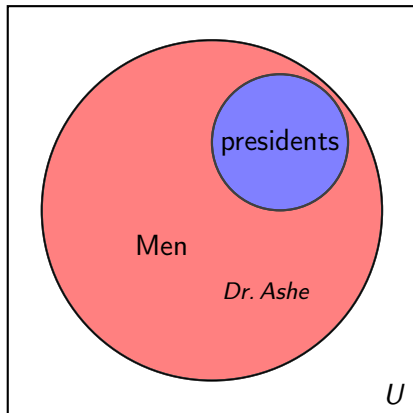
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Example

1. All presidents are men.
 2. Dr. Ashe is a man.
- ⇒ Therefore, Dr. Ashe is a US president.



Invalid Argument

- $Men \not\subseteq presidents$
- $Dr. Ashe \in Men \not\Rightarrow Dr. Ashe \in president$

Valid/Invalid Arguments

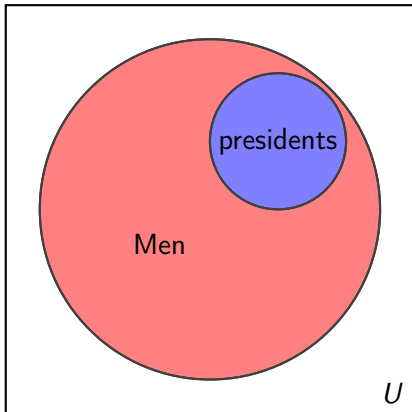
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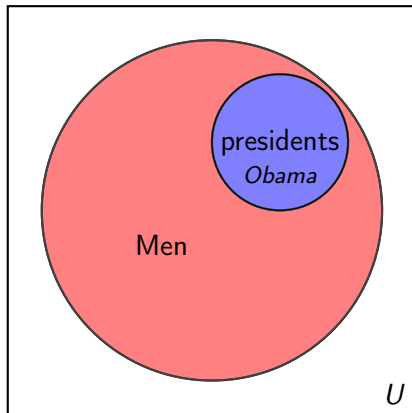
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A deductive argument is *invalid* if the premises are met, but the conclusion does *not necessarily* follow.

- Use a Venn diagram.

Example

1. All presidents are men.
2. Mr. Obama is a man.



Valid/Invalid Arguments

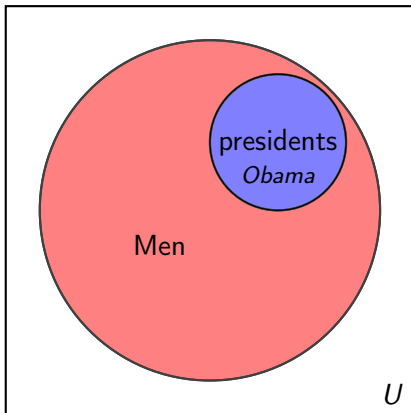
Invalid Argument

A deductive argument is *invalid* if the premises are met, but the conclusion does *not necessarily* follow.

- Use a Venn diagram.
- Correct conclusion $\nRightarrow$ valid argument.

Example

1. All presidents are men.
 2. Mr. Obama is a man.
- $\Rightarrow$ Therefore, Mr. Obama is a president.



Invalid Argument

- $Men \not\subseteq presidents$
- $x \in Men \nRightarrow x \notin president$

Valid/Invalid Arguments

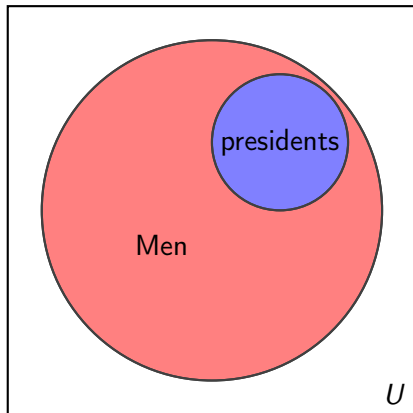
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Example

1. All US presidents are men.



Valid/Invalid Arguments

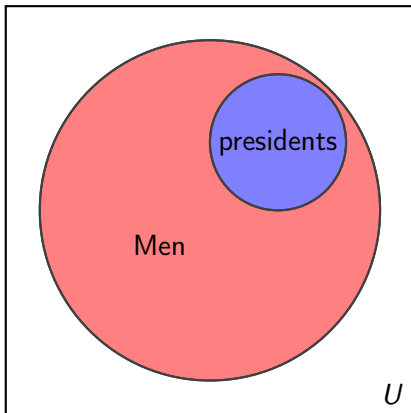
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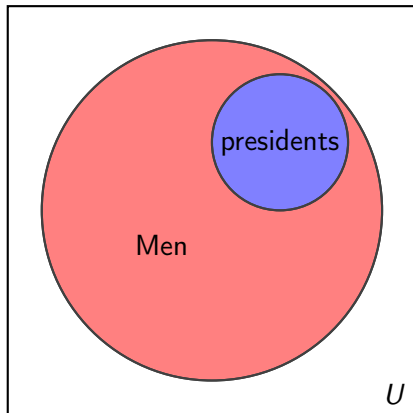
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 2. Mr. Obama is a president.
- $\Rightarrow$ Therefore, Mr. Obama is a man.



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A deductive argument is *invalid* if the premises are met, but the conclusion does *not necessarily* follow.

- Use a Venn diagram.
- Correct conclusion $\nRightarrow$ valid argument.

Example

1. All US presidents are men.
 2. Mr. Obama is a president.
- $\Rightarrow$ Therefore, Mr. Obama is a man.

Valid Argument

- $presidents \subseteq Men$
- $x \in presidents \Rightarrow x \in Men$

Problem 1.1.1. Construct a Venn diagram to determine the validity of the argument

Problem 1.1.2. Construct a Venn diagram to determine the validity of the argument

1. All Olympic gold medalists are role models.
-

Problem 1.1.3. Construct a Venn diagram to determine the validity of the argument

1. All professional wrestlers are actors.
 2. Ralph Nader is not an actor.
-

Problem 1.1.4. Construct a Venn diagram to determine the validity of the argument

1. All professional wrestlers are actors.
 2. Ralph Nader is not a professional wrestler.
-

Therefore, Ralph Nader is not an actor.

Problem 1.1.5. Construct a Venn diagram to determine the validity of the argument

1. All professional wrestlers are actors.
 2. "The Rock" is not a professional wrestler.
-

Therefore, "The Rock" is not an actor.

Inductive Arguments

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Inductive Arguments

Example

What comes next?

2, 4, 6, 8, 10, 12, __

- if the numbers are integers.
- if the numbers are on a clock.
- if we count by 2 except every 7th place is 😊.

Inductive Arguments

The conclusion in an inductive argument *is never guaranteed*.

Example

What comes next?

2, 4, 6, 8, 10, 12, __

- if the numbers are integers.
- if the numbers are on a clock.
- if we count by 2 except every 7th place is ☺.

Inductive Arguments

The conclusion in an inductive argument *is never guaranteed*.

Example

What comes next?

2, 4, 6, 8, 10, 12, __

- if the numbers are integers.
- if the numbers are on a clock.
- if we count by 2 except every 7th place is ☺.

Problem 1.1.6. Fill in the blank with what is *most likely* to be the next. Explain the pattern generated by your answer.

0, 5, 10, 15, 20, ...

Inductive Arguments

The conclusion in an inductive argument *is never guaranteed*.

Example

What comes next?

2, 4, 6, 8, 10, 12, __

- if the numbers are integers.
- if the numbers are on a clock.
- if we count by 2 except every 7th place is ☺.

Problem 1.1.7. Fill in the blank with what is *most likely* to be the next. Explain the pattern generated by your answer.

0, 2, 6, 12, 20, ...

Inductive Arguments

The conclusion in an inductive argument *is never guaranteed*.

Example

What comes next?

2, 4, 6, 8, 10, 12, __

- if the numbers are integers.
- if the numbers are on a clock.
- if we count by 2 except every 7th place is ☺.

Problem 1.1.8. Fill in the blank with what is *most likely* to be the next. Explain the pattern generated by your answer.

1, 1, 2, 3, 5, ...

Inductive Arguments

The conclusion in an inductive argument *is never guaranteed*.

Example

What comes next?

2, 4, 6, 8, 10, 12, __

- if the numbers are integers.
- if the numbers are on a clock.
- if we count by 2 except every 7th place is ☺.

Problem 1.1.9. Fill in the blank with what is *most likely* to be the next. Explain the pattern generated by your answer.

2, 3, 5, 7, 11, ...